

EDEM, FROILLAN KIM B.

Labangon, Cebu City | +63 9910522445 | froillan.edem@gmail.com
GitHub: <https://github.com/Froillan123> | LinkedIn: <https://tinyurl.com/yc6hd2nx>

PROFESSIONAL SUMMARY

Cloud/Platform and Cloud Security Engineer with hands-on experience designing and building a GitOps-driven internal developer platform on Google Cloud. Conducted an authorized cloud security assessment identifying 15+ vulnerabilities. Built a custom Go API gateway, zero-trust service-to-service IAM, and an envelope-encryption secret vault with SOC 2-aligned audit logging. Seeking cloud, platform, DevOps, or cloud security engineering roles.

EDUCATION

University of Cebu (UC) | Bachelor of Science in Information Technology | Graduated 2026

- Focus: Software Development, Cloud Computing, Database Systems
- Capstone Project: FaceOfMind — AI-Powered Mental Health Platform

TECHNICAL SKILLS

- Cloud: Google Cloud Platform (Cloud Run, Cloud Build, Cloud Functions, IAM, KMS, Load Balancing, VPC, Secret Manager)
- DevOps/GitOps: Docker, GitHub Actions, Git, Cloud Build, dynamic matrix builds, automated multi-service deployment pipelines
- Architecture: Microservices, API gateway design, dynamic routing, service registry, versioned deployments, service-to-service authentication, reverse proxying, gRPC/HTTP2, WebSocket upgrades, SSE
- Security: IAM-based access control, OIDC token generation, RSA JWT signing, AES-256-GCM envelope encryption, KMS-managed DEKs, service-isolated secret scoping, structured security audit logging
- Languages: Go, Python, C# (.NET Core), SQL, Dart, TypeScript
- Databases: PostgreSQL, Redis, MongoDB
- Observability: Latency tracking, compliance headers, health monitoring dashboards

EXPERIENCE

Cloud Security Intern (Authorized Assessment)

Philippine Real Estate Platform | Jan 2026 – Mar 2026

- Engaged by Filipino Homes to perform an authorized security assessment of cloud infrastructure, identifying 15+ vulnerabilities across data exposure, access control, database configuration, and credential management.
- Audited public S3 buckets and discovered unauthorized exposure of company financial records, customer PRC licenses, and driver's licenses.
- Flagged misconfigured security groups and backend databases on public subnets; recommended VPC private subnet and bastion host architecture to eliminate direct public exposure.
- Authored remediation reports evaluating infrastructure failures against RA 10173 (Philippine Data Privacy Act) compliance standards.
- Presented technical remediation reports to senior leadership covering database security, access controls, and cloud cost efficiency.

PROJECTS

Internal Developer Platform (FaceOfMind Infrastructure)

faceofmind.com | 2025 – 2026

GitOps Deployment Pipeline

- Built a GitOps auto-registration pipeline using GitHub Actions that scans service.yaml manifests, dynamically builds a deployment matrix, and reduces unnecessary CI/CD work by building only modified services.
- Implemented Cloud Build asynchronous image builds to avoid rate-limit quotas across concurrent service builds.
- Automated IAM binding; each deployed service grants roles/run.invoker exclusively to the API Gateway service account to enforce least-privilege access.
- Synced service and route metadata to a PostgreSQL-backed service registry, enabling dynamic route updates without gateway restarts.

Custom Go API Gateway

- Designed and built a Go-based reverse proxy gateway serving as the single public entry point for all internal microservices, scalable to handle any number of services and functions.
- Implemented longest-prefix and priority-based dynamic routing loaded from a PostgreSQL service registry, enabling hot route reloads without gateway restarts.
- Built Google OIDC ID token generation (RSA PKCS8 signing, JWT assertion, OAuth2 token exchange) with in-memory caching for authenticated service-to-service calls, eliminating repeated token exchange round-trips.
- Configured HTTP/2 transport, unbuffered flushing (FlushInterval = -1), gRPC trailer preservation, and WebSocket upgrades for real-time chat and streaming without proxy buffering delays.
- Injected latency headers (X-Gateway-Latency, X-Upstream-Latency, X-Total-Latency) on every request for end-to-end observability.

KMS-Backed Secret Vault and Envelope Encryption

- Architected a centralized secret vault on PostgreSQL with AES-256-GCM envelope encryption, where Data Encryption Keys (DEKs) are wrapped by GCP KMS to avoid storing plaintext keys at rest.
- Reduced developer bootstrap from 30+ .env API keys to two environment variables (PLATFORM_DB_URL, KEYS_KMS_ENCRYPTED_DEK) to enforce least-privilege configuration.
- Implemented service-isolated secret scoping; each microservice can only read secrets scoped to its own service_name or global to prevent lateral credential exposure.
- Built an in-memory and Redis config cache that loads decrypted secrets in ~2ms on local dev and eliminates repeated database queries in production, reducing KMS call costs and latency.
- Enforced a 90-day cryptographic key rotation policy and structured security audit logging (IP, timestamp, requested service, status) on every secret read and decrypt.